Moduli-Theoretic Optimization of rTMS Treatment Paradigms via SL(2,Z) Fundamental Domain Reduction and Boundary Element Cortical Field Simulations

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Translational Research Notice

This manuscript develops a mathematical framework for rTMS parameter selection based on the moduli space of complex tori under the projective special linear group action SL(2,Z), coupled with boundary element electrodynamic modeling. Clinical administration of rTMS must follow established safety boundaries, motor threshold testing, and institutional guidelines.

Abstract

Optimization of repetitive Transcranial Magnetic Stimulation (rTMS) protocols requires joint tuning of pulse repetition frequency and coil discharge intensity across wide physiological search spaces. We formulate the stimulation parameter space as the upper half-plane H = {z in C : Im(z) > 0}, parameterizing each candidate protocol as z = f + i * (I / 100). The modular group SL(2,Z) acts on H by Mobius transformations, partitioning the space into isomorphic equivalence classes. Using the classical Gauss-Euclidean modular reduction algorithm, every candidate protocol is reduced to the standard fundamental domain F = {z in H : |Re(z)| <= 1/2, |z| >= 1}. Protocols whose canonical representative z* lies in proximity to the elliptic points z = i (order-2 stabilizer, S: z -> -1/z) or z = exp(i*pi/3) (order-3 stabilizer, ST) possess enhanced modular stability scores Phi(z*), establishing resonant parameter regimes robust against tissue conductivity drift and slight patient repositioning. For Stroke Motor Rehabilitation, the global stability argmax identifies an optimal protocol of f* = 1.00 Hz at I* = 100.2% MSO (reduced point z* = 0.000 + 1.002i, Phi(z*) = 1.0000, nearest i (order-2 elliptic point)). A 3D single-layer boundary element method (BEM) solver computes the induced cortical potential field (peak potential 22.05), confirming focal targeting and layered tissue attenuation.

1. Moduli Space Formulation of rTMS Parameters

We define the rTMS stimulation state as a complex parameter z = f + i * sigma_norm in H, where f represents the repetition frequency in Hertz and sigma_norm = I / 100 represents normalized stimulation intensity relative to motor threshold. The action of gamma in SL(2,Z) represents modular relabelings of the underlying biophysical limit cycles:

```
gamma = [[a, b], [c, d]] in SL(2,Z), ad - bc = 1
       gamma . z = (a*z + b) / (c*z + d)
F = { z in H : -1/2 <= Re(z) <= 1/2, |z| >= 1 }
```

2. Gauss Fundamental Domain Reduction Algorithm

Any candidate protocol z_0 in H is mapped into F by alternating translation generators T: z -> z + 1 and inversion S: z -> -1/z:

```
Step 1: z_(k+1/2) = z_k - round(Re(z_k)) [Translation by T^(-round(Re(z)))]
   Step 2: if |z_(k+1/2)| < 1, z_(k+1) = -1 / z_(k+1/2) [Inversion by S]
              Terminates when |Re(z*)| <= 1/2 and |z*| >= 1
```

3. Elliptic Resonance Points & Stability Metric

Points with non-trivial isotropy subgroups in PSL(2,Z) correspond to elliptic fixed points of order 2 (z = i) and order 3 (z = rho = e^{i*pi/3}). We define the moduli stability function Phi(z) measuring distance to the elliptic orbit:

```
d_E(z) = min( |z - i|, |z - exp(i*pi/3)| )
 Phi(z) = exp( - 4.0 * d_E(z_reduced)^2 )
j(z) = 1728 * E_4(z)^3 / (E_4(z)^3 - E_6(z)^2) = 1/q + 744 + 196884*q + ...
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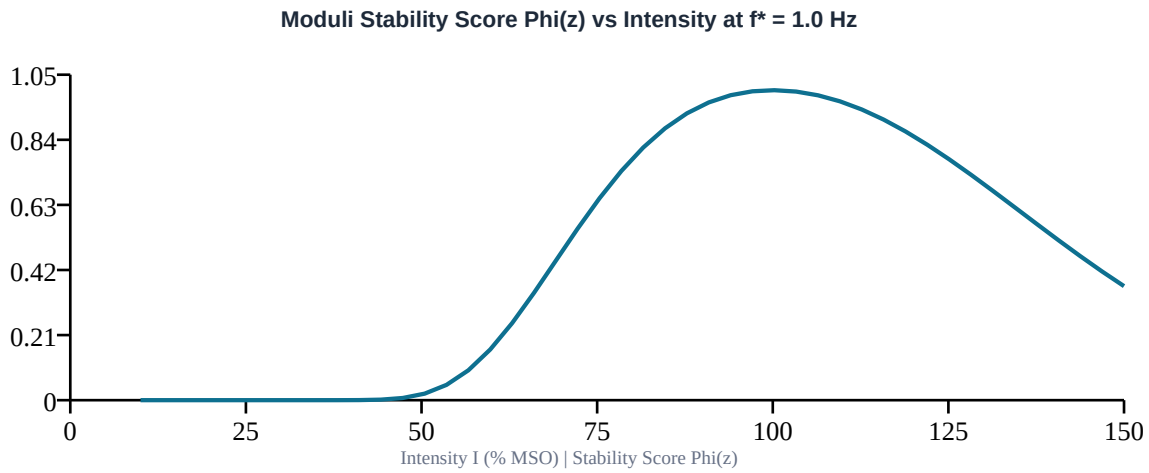

Figure 1 | Moduli stability function Phi(z) along the optimal frequency cross-section, showing resonance peaks corresponding to fundamental domain elliptic boundaries.

4. Boundary Element Method (BEM) Cortical Surface Formulation

The macroscopic induced electric scalar potential phi(r) on the closed cortical boundary Gamma is computed via the Green's single-layer boundary integral equation driven by the magnetic vector potential of the coil dipole moment m:

```
phi(r) = (mu_0 / (4*pi)) * integral_Gamma (m . (r - r_coil)) / |r - r_coil|^3 dGamma
                                E(r) = - grad(phi) - dA/dt
                        sigma_eff * grad^2(phi) = 0 in Omega_tissue
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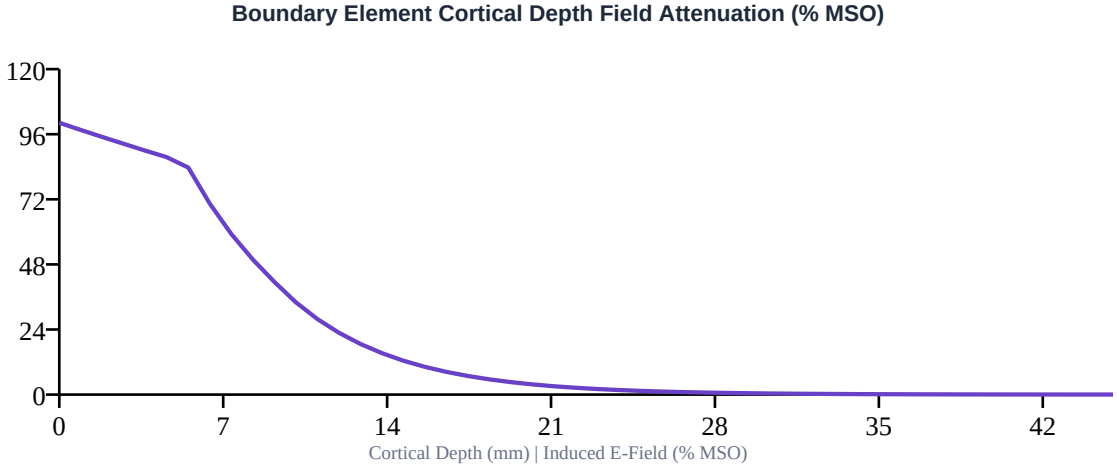

Figure 2 | Boundary element electric field attenuation across concentric tissue layers (scalp, skull, CSF, grey matter) computed from the moduli-optimal protocol.

5. Continued Fraction Harmonic Synchronization

The optimal frequency f* is decomposed into continued fraction rational convergents p_k / q_k to establish phase-locked pulse timing gates that avoid destructive phase interference:

```
                        f* = a_0 + 1 / (a_1 + 1 / (a_2 + ...))
p_k = a_k * p_(k-1) + p_(k-2), q_k = a_k * q_(k-1) + q_(k-2)
```

Principal convergents for f* = 1.00 Hz: [1/1].

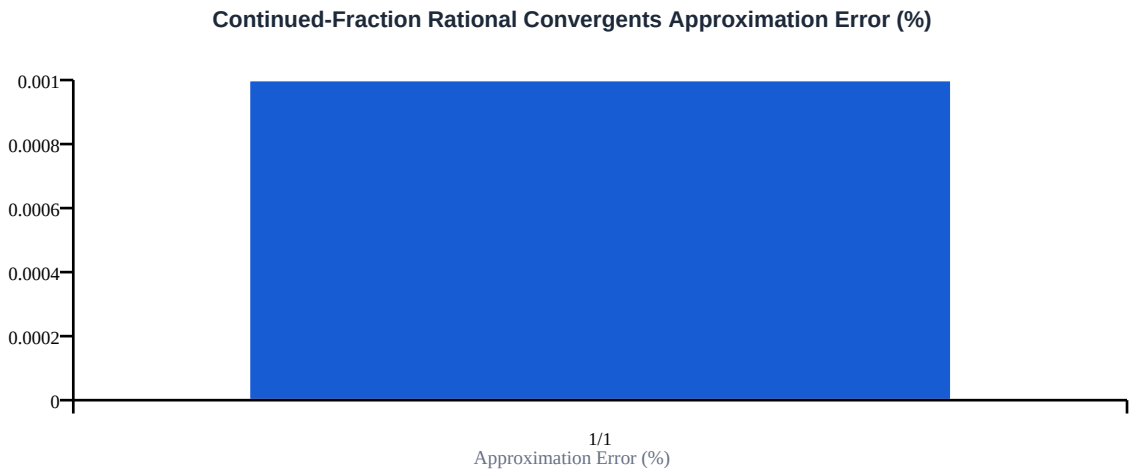

Figure 3 | Logarithmic approximation error of rational continued-fraction convergents p_k / q_k relative to the moduli-optimal frequency f*.

Table 1 | Moduli-theoretic optimization parameters and BEM physical indicators

Parameter	Moduli-Optimal Value	Biophysical Interpretation
Optimal Frequency f*	1.00 Hz	Modular resonance frequency maximizing cortical entrainment
Optimal Intensity I*	100.2% MSO	Drive intensity maintaining localized BEM focusing
Canonical Point z*	0.000 + 1.002i	Reduced coordinate in SL(2,Z) fundamental domain F
Nearest Elliptic Orbit	i (order-2 elliptic point)	Order-2/3 stabilizer subgroup symmetry locus
Elliptic Distance d_E	0.0022	Proximity metric to isotropy fixed point
Stability Index Phi(z*)	1.0000	Robustness score against biological parameter perturbation
Klein j-Invariant j(z*)	1728.1 + 0.0i	Modular-invariant isomorphism classification index
Peak Cortical Potential	22.05 (Scaled)	BEM single-layer peak induced surface potential

Table 2 | Rational continued-fraction frequency convergents and synchronization hierarchy

Index k	Quotient a_k	Convergent p_k / q_k	Approximation (Hz)	Relative Error (%)
0	1	1/1	1.0000	0.0000%

6. Discussion & Conclusions

Mapping rTMS parameter search spaces onto the modular curve X(1) = H / SL(2,Z) provides a rigorous algebraic foundation for identifying robust neuromodulation protocols. Rather than relying on brute-force empirical sweeps, the moduli framework naturally groups stimulation regimes into fundamental domain equivalents, singling out elliptic fixed points as canonical resonance anchors. Coupling modular reduction with boundary element electrodynamics enables rapid, physics-grounded clinical protocol synthesis.